

Introduction to AI

SUBJECT

Computer Science

GRADE LEVEL

Grade 3

DURATION

30 minutes

Learning Objectives

- 1 Students will be able to define Artificial Intelligence (AI) in simple terms.
- 2 Students will be able to identify AI Agents and their functions.
- 3 Students will be able to recognize various AI Applications in daily life.
- 4 Students will be able to differentiate among ChatGPT, Gemini, and Grok as AI tools.

Materials Needed

- Whiteboard or projector
- Markers or pens
- Computer with internet access (for teacher demonstration)
- Handout with images of AI Agents and Applications (1 per student)
- Worksheet with fill-in-the-blanks and matching exercises (1 per student)
- Chart paper or large sticky notes
- Markers for chart paper



Introduction / Hook

(5 minutes)

- * **Engage (2 minutes):** Begin by asking students: "Have you ever heard of robots that can talk or play games?" Allow a few students to share their experiences.
- * **Introduce (3 minutes):** Explain that these robots and smart devices use something called Artificial Intelligence, or AI. Briefly define AI as computers that can think and learn like humans.



Direct Instruction

(15 minutes)

- * **Defining AI (3 minutes):**
 - * Clearly define AI: "Artificial Intelligence means making computers smart so they can do things that usually need a human brain, like understanding words or solving problems."
 - * Provide simple examples: "Think of a smart speaker that can answer your questions or a game that learns how you play and gets harder."
- * **AI Agents (5 minutes):**
 - * Introduce the concept of AI Agents: "AI Agents are the smart parts of machines or programs that can do things for us. They sense their environment and then act to achieve a goal."
 - * Examples: Voice assistants like Siri or Alexa, chatbots on websites, self-driving cars.
 - * Show images of AI Agents from the handout. Discuss what each agent does (e.g., voice assistant answers questions, chatbot helps with online shopping).
- * **AI Applications (5 minutes):**
 - * Explain that AI is used in many different things we use every day: "AI is all around us! Let's look at where we can find AI helping us."
 - * Examples: Spam filters in email, recommendation systems on Netflix, facial recognition on phones.
 - * Show images of AI Applications from the handout. Discuss how each application helps people (e.g., spam filter protects from junk mail, Netflix suggests shows).
- * **Brief introduction to ChatGPT, Gemini, and Grok (2 minutes):** Explain that these are examples of large language models, which are types of AI that can generate human-like text. Use simplified terms and avoid overwhelming details. Explain that they can answer questions, write stories, and more. Emphasize that these are just examples, and there are many other AI tools.



Guided Practice

(5 minutes)

* **Think-Pair-Share Activity (5 minutes):**

* Pose the question: "What is one thing you would like an AI Agent to do for you?"

* **Think (1 minute):** Students think individually about their answer.

* **Pair (2 minutes):** Students pair up with a classmate and share their answers.

* **Share (2 minutes):** Call on a few pairs to share their ideas with the whole class. Write some of the ideas on the chart paper.



Independent Practice

(0 minutes)

* This lesson does not include independent practice due to time constraints. The worksheet is assigned as homework instead.



Closure

(3 minutes)

* **Review (2 minutes):** Briefly review the key concepts: What is AI? What are AI Agents? Where do we find AI Applications?

* **Question (1 minute):** Ask students: "Can you think of one new thing you learned about AI today?" Allow a few students to share.



Assessment

(Formative)

* **Observation:** Observe student participation in the Think-Pair-Share activity and during class discussions.

* **Questions:** Ask probing questions throughout the lesson to gauge understanding, such as: "Why is AI useful?", "Can you give me another example of an AI Agent?", "How does AI help us in our daily lives?"



Differentiation

- * **Struggling Learners:** Provide simplified explanations and more visual aids. Offer one-on-one support during activities. Allow them to draw pictures to represent AI concepts.
- * **On-Level Students:** Encourage them to elaborate on their answers and provide more examples. Challenge them to identify AI applications they haven't heard of before.
- * **Advanced Learners:** Ask them to research a specific AI application (e.g., AI in healthcare) and share their findings with the class in the next lesson (as an extension of the lesson).



Accommodations

- * **ELL Students:** Provide visual aids and simplified language. Allow them to respond in their native language (if possible) and translate later. Partner them with a bilingual student.
- * **IEP/504 Students:** Follow their individual education plans (IEPs) or 504 plans. Provide preferential seating, extended time, or modified assignments as needed. Break down tasks into smaller, more manageable steps.



Extensions

- * **AI Ethics Discussion:** Briefly introduce the idea that AI can be used for good and bad purposes. Ask students to brainstorm examples of each. (This could be a starting point for a future lesson.)
- * **Research Project:** Assign interested students to research a specific AI tool or application and present their findings to the class.



Homework / Follow-up

- * **Worksheet:** Complete the worksheet with fill-in-the-blanks and matching exercises to reinforce the concepts learned in class. (Provide as printed sheet)



Teacher Reflection Notes

- How engaged were the students with the topic of AI?
- Were the examples of AI Agents and Applications relevant to their lives?
- What adjustments could be made to the lesson to better meet the needs of all students?
- Did the time allocation for each activity work effectively?
- How could the introduction to ChatGPT, Gemini, and Grok be improved for this age group?
- Did the formative assessments accurately gauge student understanding?